EXP NO: 7 DATE: 3-03-2026

FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS:

AIM:

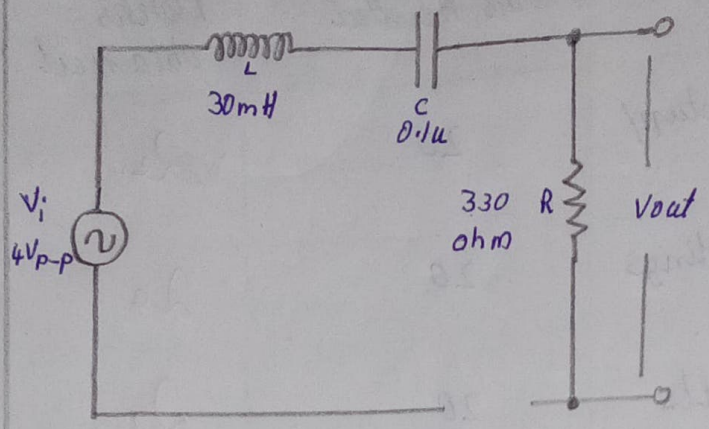
To study the behavior of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwith and Q of the given RLC circuit using Multisim Simulator.

APPARATUS REQUIRED:

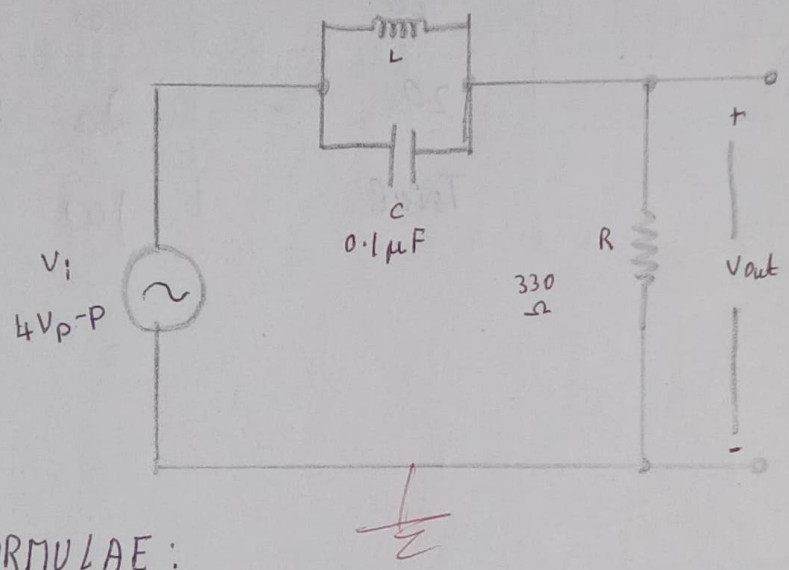
S.NO	Components	Specification	Quantity
1	CRO	10 Mhz	1
2	DIB	-	1
3	Capacitors	0.1 μf	1
4.	Resistors	330 Ω	1
5.	Bread Board	--	--
6.	Connecting wires	--	--
7.	PC with Multisim Simulator	--	--

CIRCUIT DIAGRAM:

SERIES RLC circuit:



Parallel RLC circuit:



FORMULAE:

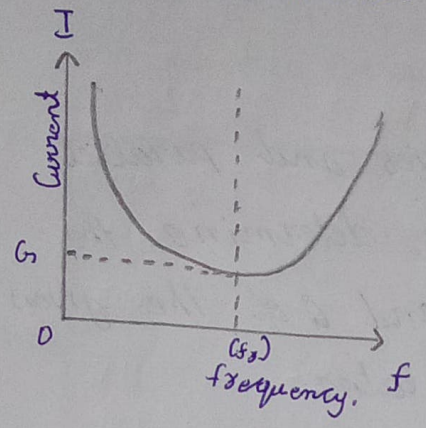
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW = f_2 - f_1$$

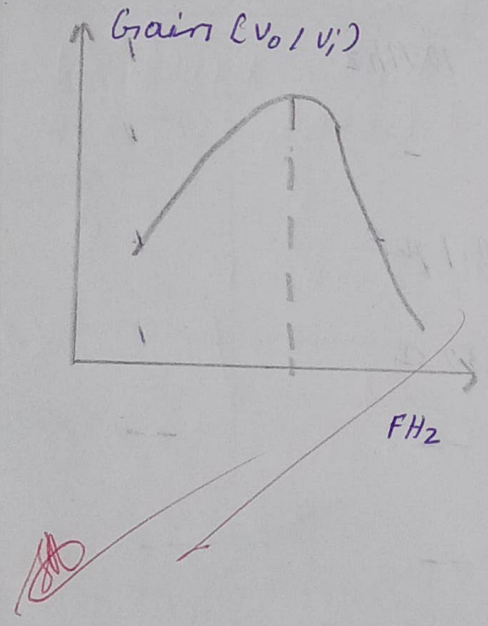
$$Q = f_0 / BW$$

MODEL GRAPH:

PARALLEL RESONANCE:



SERIES RESONANCE:



PROCEDURE:

1. Construct circuit shown in figure 1 on breadboard.

2. Connect CRO ch1 to input and ch2 to output.

3. Set the input Voltage to 4 V_{p-p}.

4. Vary the frequency from 500Hz to 3kHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage V_m is the resonance frequency.

5) Calculate the bandwidth by subtracting the 2 cut-off frequencies.

6) Calculate the Q factor

7) Repeat steps 1 through 7 for circuit shown in Figure 2.

TABULATION :

$V_{in} = 4 V_{p-p}$

SERIES RESONANCE			PARALLEL RESONANCE		
Frequency (Hz)	Output Voltage (V)	Output Current (mA)	Frequency (Hz)	Output Voltage (V)	Output Current (mA)
500	1.2	2	500	1.2	9
700	2.0	4	700	2.8	7
900	3.5	7	900	4.0	5
(Resonance) 1000	5.0	10	(Resonance) 1000	5.0	2 (min)
1100	3.6	7	1100	4.1	5

Calculation:

$L = 100\,mH = 0.1\,H$

$C = 0.25\,\mu F = 2.5 \times 10^{-6} F$

$$F_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25 \times 10^{-6})}} = \frac{1}{2\pi\sqrt{1.58 \times 10^{-4}}}$$

$$F_0 = \frac{1}{9.92 \times 10^{-4}} = 1000\ Hz$$

Maximum current (series) and minimum current (Parallel) occur at 1000 Hz.

$$BW = \frac{R}{2\pi L} = \frac{330}{2 \times 3.14 \times 0.1} = \frac{330}{0.628} = 525.4$$

$$Q = \frac{f_0}{BW} = \frac{1000}{525.4} = 1.903$$

MARKS SPLIT-UP:

SI. NO.	Description	Marks Allotted	Marks obtained
1	Experimental Setup / Connections	20	20
2	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
			14

RESULT:

Thus the phenomenon of resonance in RLC circuit was studied and the following were determined using Multisim Simulator.

Series RLC circuit:	Parallel RLC circuit:
Resonant frequency = 1000	Resonant frequency = 1000
Bandwidth = 525.4	Bandwidth = 525.4
Q factor = 1.903	Q factor = 1.903